

Individual Evaluation Report

for the World-Wide Open Proficiency Test IAEA-TERC-2022-01 Part II

Individual Evaluation Report for Laboratory Nr. 8

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Evaluation Tables for Labcode 8. (Values and uncertainties expressed in Bq/kg)

Evaluation Result Table for Sample 1

Sample Code	Analyte	Target Value	Target Unc.	MARB	Rep. Value	Rep. Unc	Rel. Bias	Robust SD	Z-Score	Accuracy	P	Precision	Final Score
1	Co-60	17.7	1.1	20 %	17.7	1.4	0.00 %	1.1	0.00	A	10.06	A	A
1	Cs-134	15.9	1	20 %	16.0	1.2	0.63 %	1	0.10	A	9.79	A	A
1	Cs-137	24.2	1.5	20 %	24.3	1.5	0.41 %	1.5	0.07	A	8.75	A	A
1	Pb-210	31.3	1.9	30 %	31.6	2.0	0.96 %	1.9	0.16	A	8.77	A	A
1	Pb-210	31.3	1.9	30 %	32.3	3.2	3.19 %	1.9	0.53	A	11.62	A	A
1	Po-210	31.6	1.9	30 %	30.9	1.8	-2.22 %	1.9	0.37	A	8.37	A	A
1	Sr-90	26.4	1.6	30 %	27.0	1.4	2.27 %	1.6	0.38	A	7.98	A	A

Evaluation Result Table for Sample 2

Sample Code	Analyte	Target Value	Target Unc.	MARB	Rep. Value	Rep. Unc	Rel. Bias	Robust SD	Z-Score	Accuracy	P	Precision	Final Score
2	Am-241	10.1	0.6	30 %	10.2	1.0	0.99 %	0.6	0.17	A	11.46	A	A
2	Cs-137	8.36	0.5	20 %	8.24	0.56	-1.44 %	0.5	0.24	A	9.05	A	A
2	H-3	10.8	0.6	30 %	11.1	0.7	2.78 %	0.6	0.50	A	8.40	A	A
2	Pb-210	5.2	0.31	30 %	5.08	0.38	-2.31 %	0.31	0.39	A	9.57	A	A
2	Po-210	5.29	0.32	30 %	4.89	0.34	-7.56 %	0.32	1.25	A	9.22	A	A
2	Sr-90	7.42	0.45	30 %	7.35	0.4	-0.94 %	0.45	0.16	A	8.15	A	A

Evaluation Result Table for Sample 3

Sample Code	Analyte	Target Value	Target Unc.	MARB	Rep. Value	Rep. Unc	Rel. Bias	Robust SD	Z-Score	Accuracy	P	Precision	Final Score
3	Am-241	0.902	0.054	30 %	0.96	0.15	6.43 %	0.054	1.07	A	16.73	A	A
3	Cs-134	12.1	0.7	20 %	12.0	1.0	-0.83 %	0.7	0.14	A	10.14	A	A
3	Cs-137	22.6	1.4	20 %	23.2	1.2	2.65 %	1.4	0.43	A	8.07	A	A

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Values and uncertainties for Sample 5

Evaluation Result Table for Sample 5

Sample Code	Analyte	Target Value	Target Unc.	MARB	Rep. Value	Rep. Unc	Rel. Bias	Robust SD	Z-Score	Accuracy	P	Precision	Final Score
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Intercomparison Parameter Evaluation: (Units specified in Part I)

Sample Code	Analyte	Robust Mean	Robust SD	Rep. Value	Rep. Unc	Z-Score	Z-Score Evaluation
1	gross_alpha	23.98	7.78	27.7	1.4	0.48	A
1	gross_beta	124.75	29.46	126	5	0.04	A
2	gross_alpha	12.72	3.73	14.8	0.9	0.56	A
2	gross_beta	28.94	6.35	27.9	1.4	0.16	A
3	gross_alpha	8.93	3.04	8.89	0.53	0.01	A
3	gross_beta	27.63	6.78	27.0	1.1	0.09	A
5	gross_alpha	0.081	0.052	0.10	0.01	0.37	A
5	gross_beta	0.158	0.084	0.17	0.02	0.14	A
7	Am-241	78.89	7.59	81.5	8.2	0.34	A
7	Ba-133	62.13	8.76	66.7	6.4	0.52	A
7	Co-60	123.96	8.12	127	11	0.37	A
7	Cs-134	39.8	3.85	42.8	3.6	0.78	A
7	Cs-137	83.2	4.79	84.8	6.1	0.33	A
7	K-40	449.74	31.25	443	32	0.22	A

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The analytes listed in the table below have been identified but are not present in the samples (false positive):

Sample Code	Analyte	Reported Value
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